Rebuttal Draft (overall)

> 1. Insufficient ablations to isolate the benefits of variable-length grouping.

We conduct experiments on training-based spatially local 2D pooling and here are the results on LLaVA 1.5:

method	TextVQA	OCRBench	OCRBenchv2	DocVQA	ChartQAPro	Avg. Ret. (OCR)
LLaVA-1.5-7B	58.16	18.5	23.8	25.61	12.04	100%
4x compression						
fixed pooling	55.74	15.7	20.9	18.78	**11.10**	86.80%
fixed (2D)	53.37	15.7	**21.9**	18.60	10.67	85.97%
VLVLM (ours)	**55.84**	**16.5**	20.9	**20.39**	10.91	**88.65%**
8x compression						
fixed pooling	54.05	14.3	19.5	16.00	10.81	80.88%
fixed (2D)	51.54	13.6	**20.4**	16.30	9.39	77.89%
VLVLM (ours)	**55.05**	**15.6**	20.1	**17.59**	**10.94**	**84.59%**

The results demonstrate that 2D fixed pooling performs similarly as 1D fixed pooling, and our VLVLM maintains the best OCR-heavy task performance.

> 2. limited coverage of competitive baselines

In addition to *prompt-agnostic* PruMerge and PruneSID, we also conduct experiments with one *prompt-aware* training-free baseline, SparseVLM. Here are the results on LLaVA 1.5:

method	TextVQA	OCRBench	OCRBenchv2	DocVQA	ChartQAPro	Avg. Ret. (OCR)
LLaVA-1.5-7B	58.16	18.5	23.8	25.61	12.04	100%
4x compression						
SparseVLM	55.10	15.2	**21.3**	17.33	**11.46**	85.85%
VLVLM (ours)	**55.84**	**16.5**	20.9	**20.39**	10.91	**88.65%**
8x compression						
SparseVLM	52.93	11.0	18.2	11.62	**10.98**	72.70%

method	TextVQA	OCRBench	OCRBenchv2	DocVQA	ChartQAPro	Avg. Ret. (OCR)
VLVLM (ours)	**55.05**	**15.6**	**20.1**	**17.59**	10.94	**84.59%**

Note that our VLVLM outperforms SparseVLM method, where we observe significant gains under aggressive compression budget (8x).

> 3. benchmark-level latency measures on Qwen3VL

We report LLM prefill latency in miliseconds for Qwen3VL on multiple benchmarks as follows. Our method significantly reduces LLM prefill latency and achieves similar speedup as other state-of-the-art methods like PruneSID.

method	OCRBench	TextVQA	ChartQAPro
Qwen3VL	92.02	69.50	131.79
4x compression			
PruneSID	57.77	53.54	63.62
VLVLM (ours)	58.45	54.00	63.02

> 4. Weak benchmark performance.

In addition to five benchmarks in the main text, we also evaluate our method VLVLM against baselines on a more recent, real-world, high-resolution benchmark - MME-RealWorld:

Model	monitoring	OCR	Diagram and Table	Autonomous Driving	Remote Sensing	Overall	Avg. Ret.
LLaVA	19.22	43.62	20.21	26.89	22.74	28.09	100%
4x compression							
PruMerge	19.31	42.65	20.27	26.94	22.63	27.87	99.22%
PruneSID	19.03	43.14	20.61	26.64	**28.09**	28.12	100.11%
fixed pooling	**21.95**	44.16	20.43	26.20	24.48	28.69	102.14%
VLVLM (ours)	20.26	**48.24**	**21.79**	**27.49**	26.86	**30.77**	**109.54%**
8x compression							
PruMerge	19.31	42.46	20.27	26.97	22.66	27.82	99.04%
PruneSID	19.31	42.89	20.49	26.64	22.74	28.03	99.79%
fixed pooling	20.95	**47.06**	20.89	27.35	26.57	30.20	107.51%
VLVLM (ours)	**21.86**	46.95	**21.44**	**27.43**	**26.97**	**30.44**	**108.37%**

VLVLM outperforms other baselines such as PruMerge, PruneSID, and fixed pooling, under both moderate (i.e., 4$\times$) and aggressive (i.e., 8$\times$) compression budget, especially on tasks subsets like "OCR" and "Diagram and Table".

> 5. robustness of the incremental gains.

For all experiments, we already apply McNemar's test to assess whether the observed performance improvements over the baselines are statistically significant to measure the robustness of the incremental gains over trained fixed pooling. We reveal the *raw test statistics* and corresponding *p-values* for LLaVA 1.5 experiment as a more detailed example:

method	**TextVQA**	**OCRBench**	**OCRBenchv2**	**DocVQA**	**ChartQAPro**
LLaVA-1.5-7B	58.16	18.5	23.8	25.61	12.04
4x compression					
fixed pooling	55.74	15.7	20.9	18.78	**11.10**
VLVLM (ours)	**55.84**	**16.5**	20.9	**20.39**	10.91
McNemar's test	0.0733	2.127	0.237	30.30	0.179
p-value	0.786	0.145	0.626	***0.000***	0.672
8x compression					
fixed pooling	54.05	14.3	19.5	16.00	10.81
VLVLM (ours)	**55.05**	**15.6**	20.1	**17.59**	**10.94**
McNemar's test	1.481	3.25	0.575	32.397	0.455
p-value	0.223	0.071	0.448	***0.000***	0.499

On benchmark DocVQA in particular, our method VLVLM's performance gain is consistently statistically significant.

> 6. incomplete comparisons with training-based methods.

Most mentioned training-based baselines have their own limitations. *TokenPacker* is an alternative to the MLP projector in VLM architectures such as LLaVA and QwenVL. It is meant to replace the original projector in the VLM, which does not share the same purpose of our work. *LearnPruner*, *MetaCompress*, and *EvoComp* do not have open-sourced code implementation or artifacts, meaning that their results cannot be reproduced.